

Electric Circuits



*Nodal Analysis: Circuits with
Current & Voltage Sources*

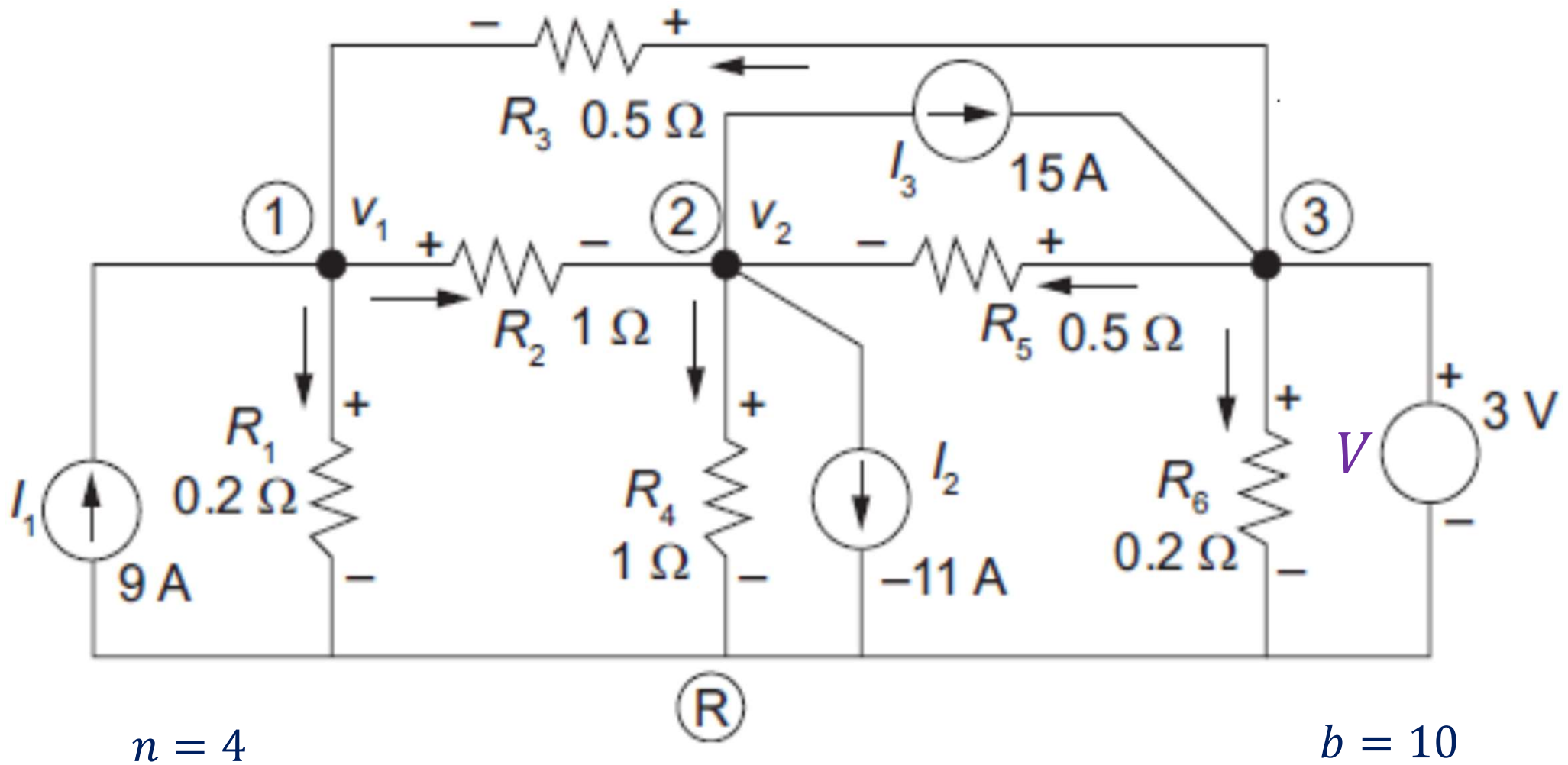
Tentative Lecture Plan

L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem		
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	FRI, 26 AUG	MON, 28 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	TUE, 29 AUG	FRI, 04 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	TUE, 05 SEP	FRI, 08 SEP	Series RL and RC Circuits		
9	SAT, 09 SEP	MON, 11 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	TUE, 12 SEP	FRI, 15 SEP	Concept of Phasor, AC Circuits, AC Power	R	SAT, 16 SEP
WED, 20 SEP: MidSem : Lectures 1 – 10					

Presence of Independent Voltage Source (IVS)

- Between A Node and Reference Node
- Between Two Nodes
- Connected in Series with some Resistor

IVS between a Node and Reference Node



IVS between a Node and Reference Node

KCL at Node 1

$$-\frac{V_1}{R_1} - \left(\frac{V_1 - V_2}{R_2} \right) + \frac{V - V_1}{R_3} + I_1 = 0$$

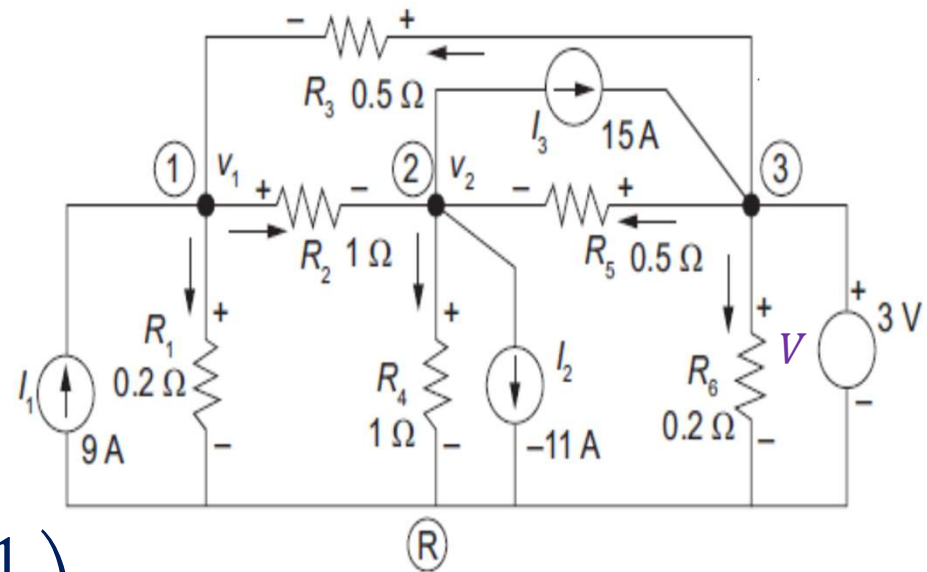


$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) - V_2 \left(\frac{1}{R_2} \right) = I_1 + V \left(\frac{1}{R_3} \right)$$



$$V_1(G_1 + G_2 + G_3) - V_2(G_2) = I_1 + V(G_3)$$

$$V_3 = V$$



IVS between a Node and Reference Node

KCL at Node 2

$$\frac{V_1 - V_2}{R_2} - \frac{V_2}{R_4} + \frac{V - V_2}{R_5} - I_2 - I_3 = 0$$

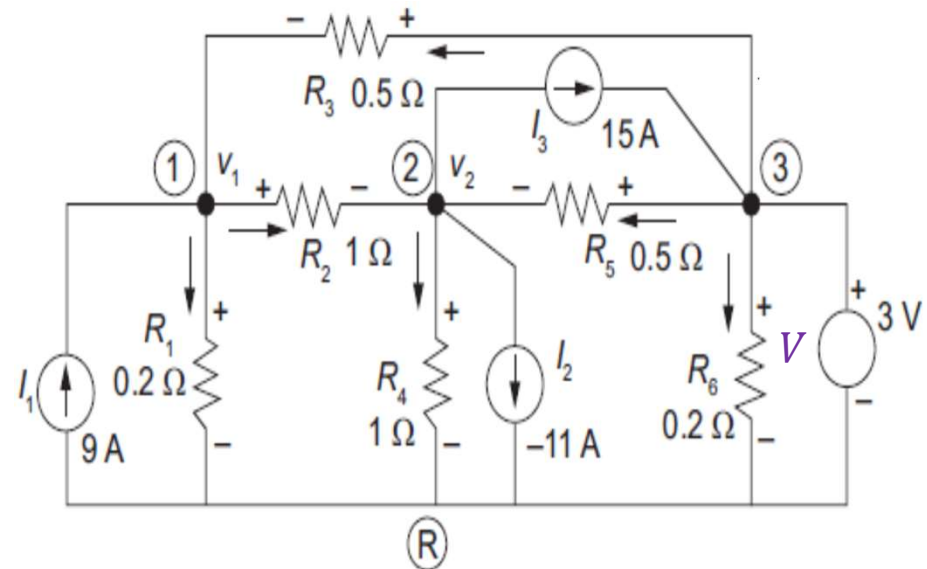


$$-V_1 \left(\frac{1}{R_2} \right) + V_2 \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} \right) - V \left(\frac{1}{R_5} \right) = -I_2 - I_3$$



$$-V_1(G_2) + V_2(G_2 + G_4 + G_5) = -I_2 - I_3 + V(G_5)$$

$$V_3 = V$$



IVS between a Node and Reference Node

$$V_1(G_1 + G_2 + G_3) - V_2(G_2) = I_1 + V(G_3)$$

$$-V_1(G_2) + V_2(G_2 + G_4 + G_5) = -I_2 - I_3 + V(G_5)$$

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 \\ -G_2 & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & G_3 \\ 0 & -1 & -1 & G_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V \end{bmatrix}$$

$$[Y]_{(n-1-n_{ivs}) \times (n-1-n_{ivs})} [V]_{(n-1-n_{ivs}) \times 1} = [C]_{(n-1-n_{ivs}) \times (n_{ics}+n_{ivs})} [U]_{(n_{ics}+n_{ivs}) \times 1}$$

n_{ics} : Number of Independent Current Sources

n_{ivs} : Number of Independent Voltage Sources

IVS between a Node and Reference Node

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 \\ -G_2 & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & G_3 \\ 0 & -1 & -1 & G_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V \end{bmatrix}$$

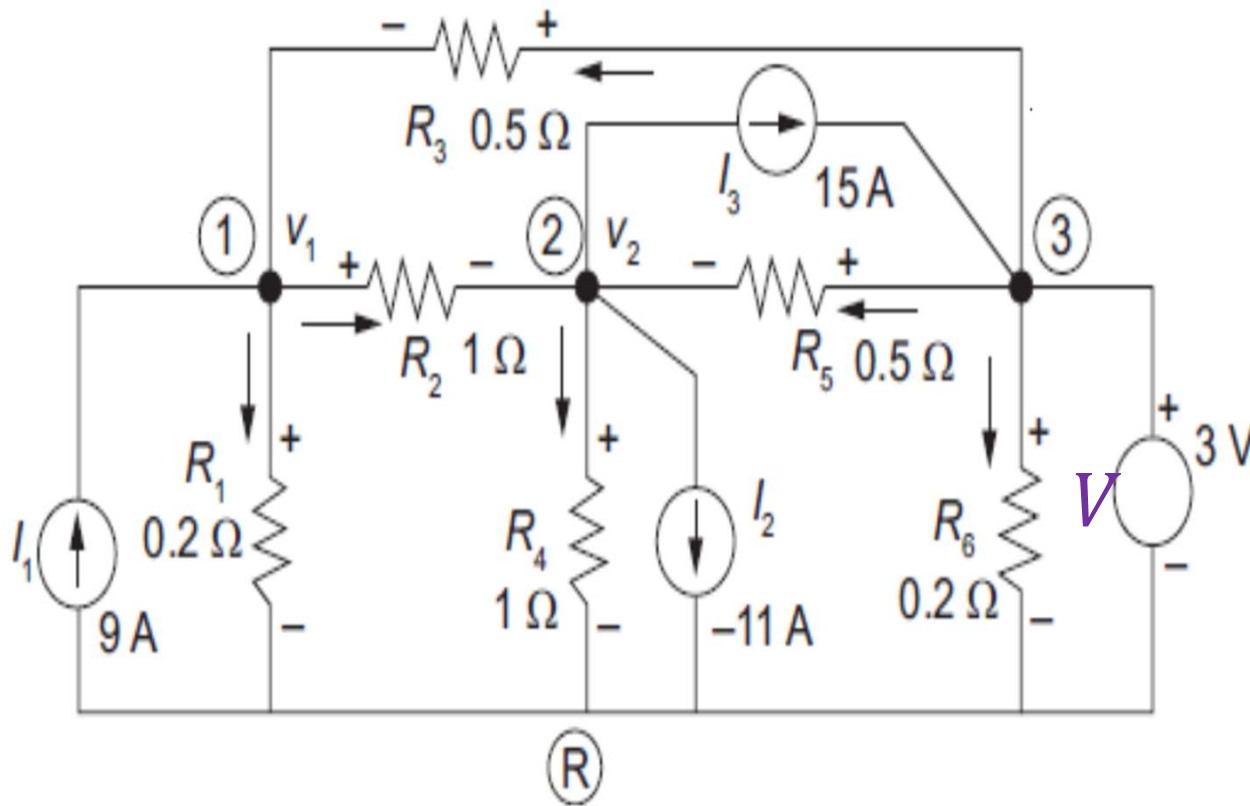


$$\begin{bmatrix} 8 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 15 \\ 2 \end{bmatrix}$$



$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

IVS between a Node and Reference Node

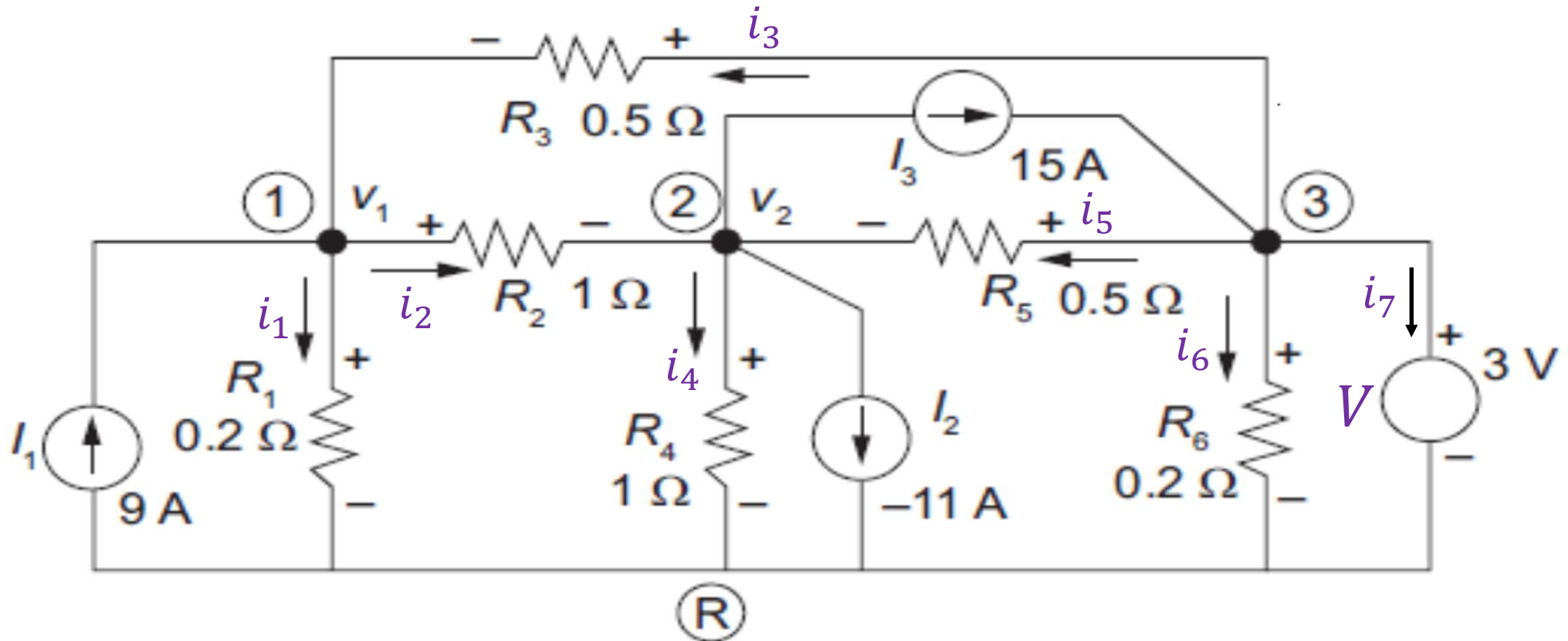


$$n = 4 \quad b = 10$$

$$V_1 = 2 \quad V_2 = 1$$

How many Branch Current Variables are to be Solved?

IVS between a Node and Reference Node



$$i_1 = 10\text{ A}$$

$$i_3 = 2\text{ A}$$

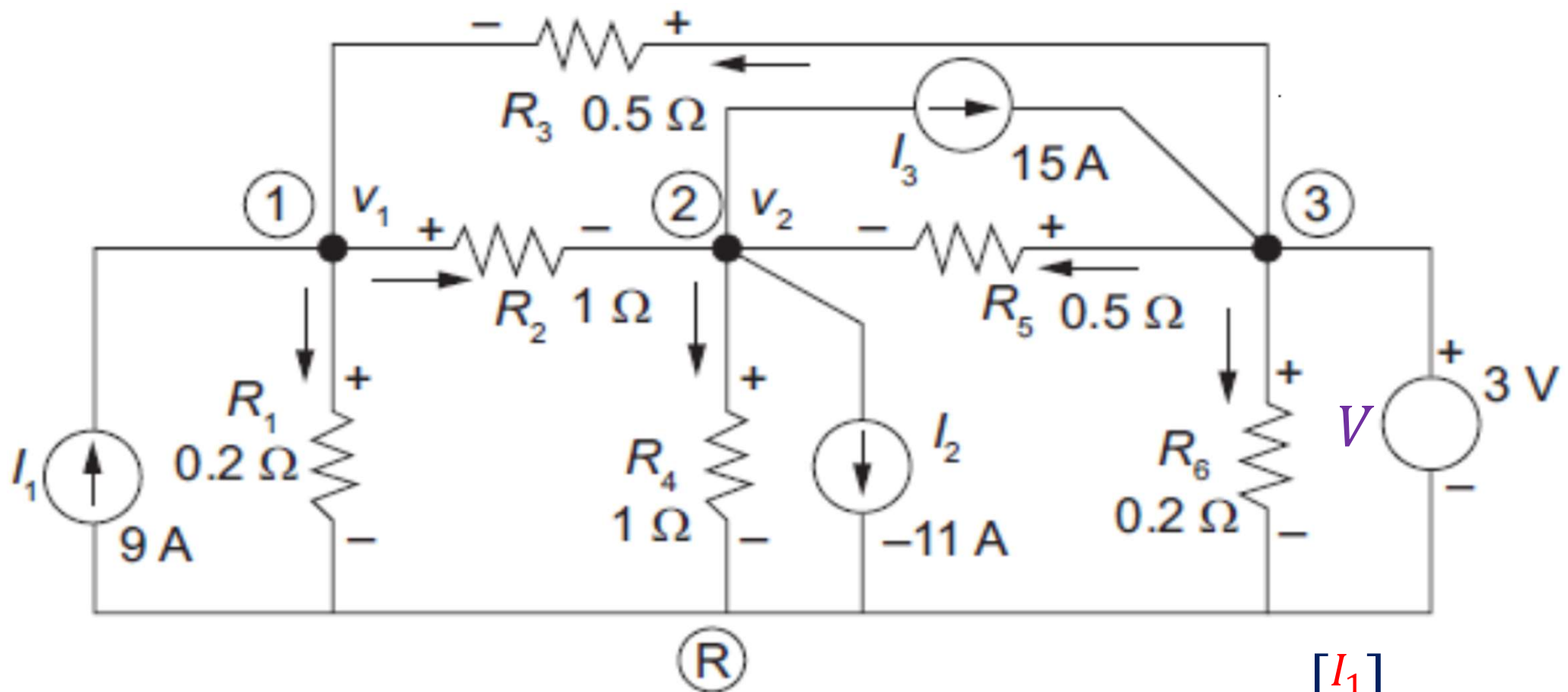
$$i_5 = 4\text{ A}$$

$$i_7 = -6\text{ A}$$

$$i_2 = 1\text{ A}$$

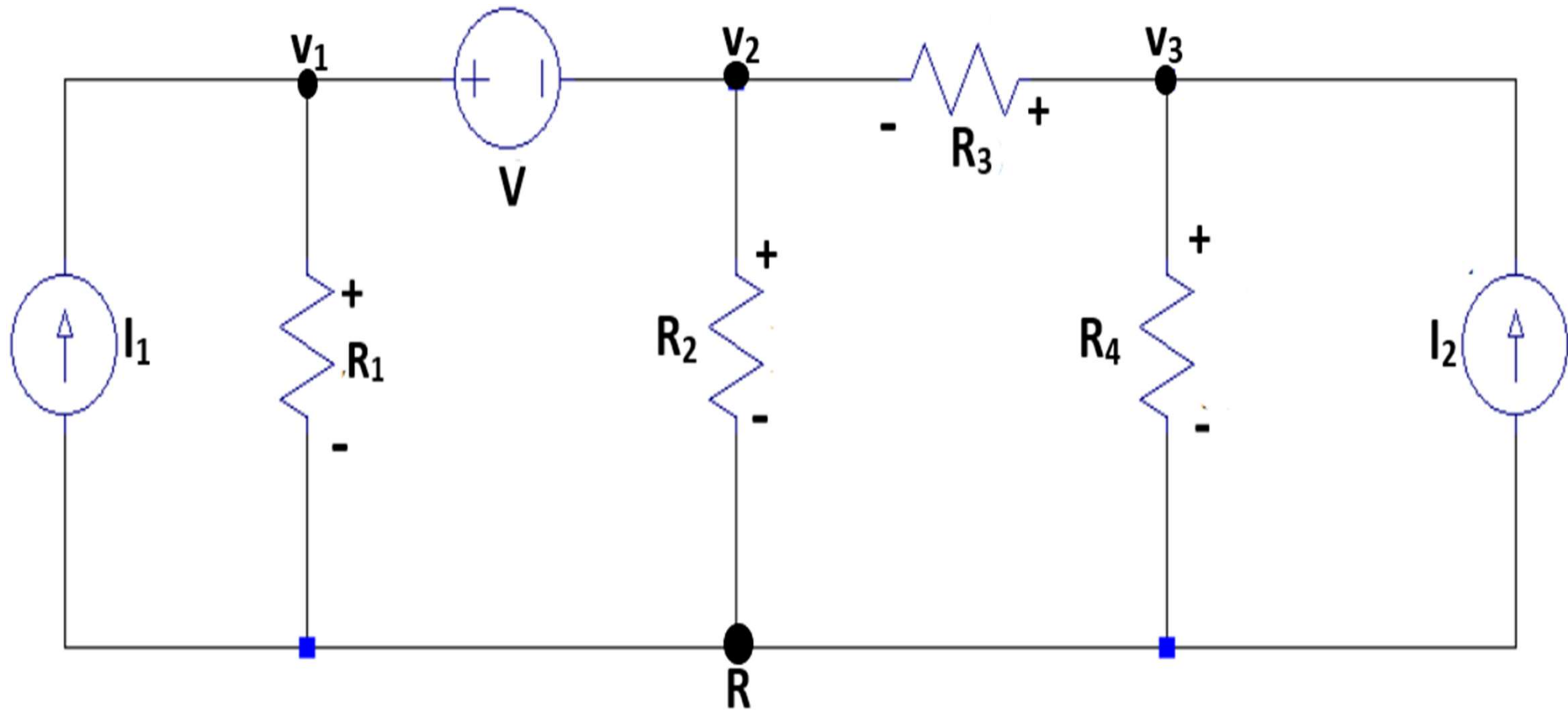
$$i_4 = 1\text{ A}$$

$$i_6 = 15\text{ A}$$

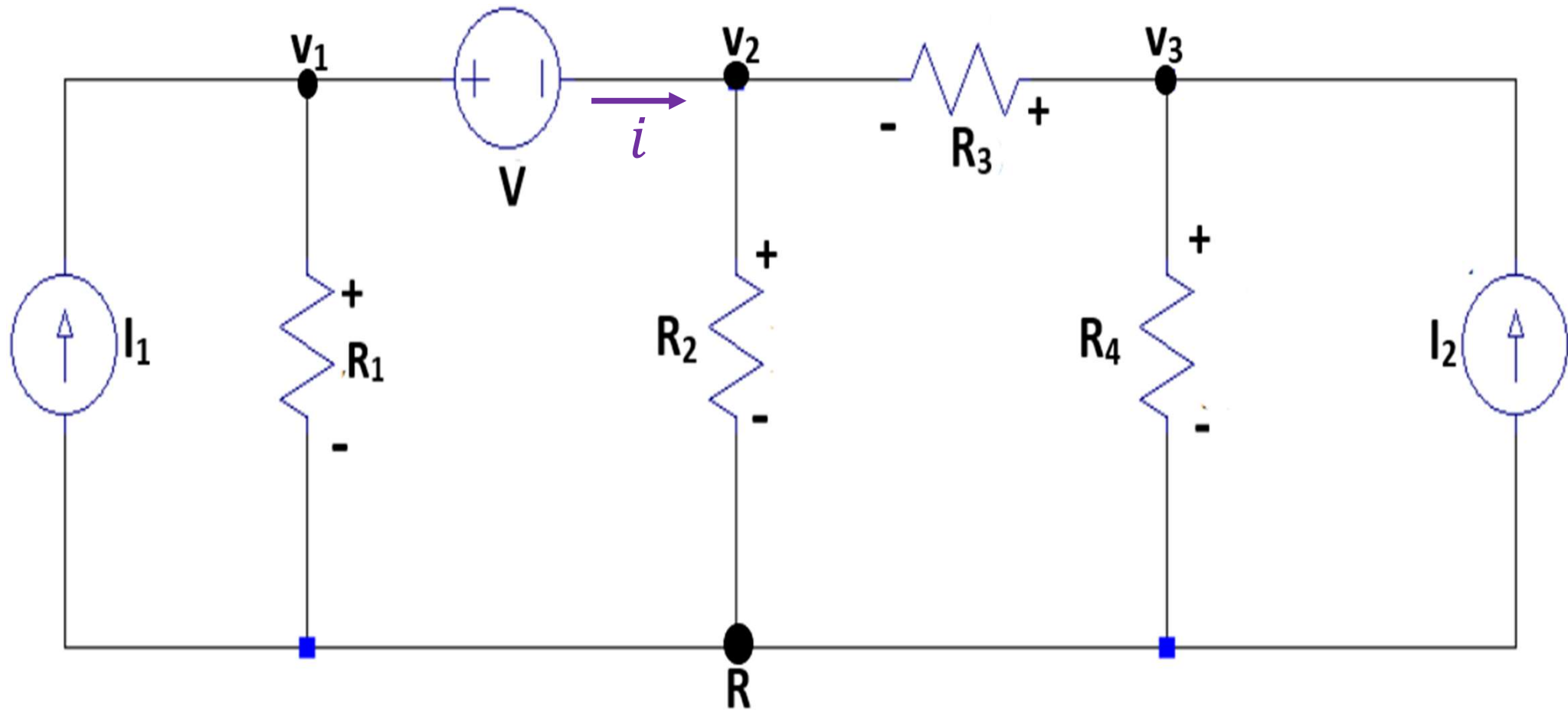


$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 \\ -G_2 & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & G_3 \\ 0 & -1 & -1 & G_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V \end{bmatrix}$$

IVS between Two Nodes



IVS between Two Nodes



IVS between Two Nodes

KCL at Node **1**

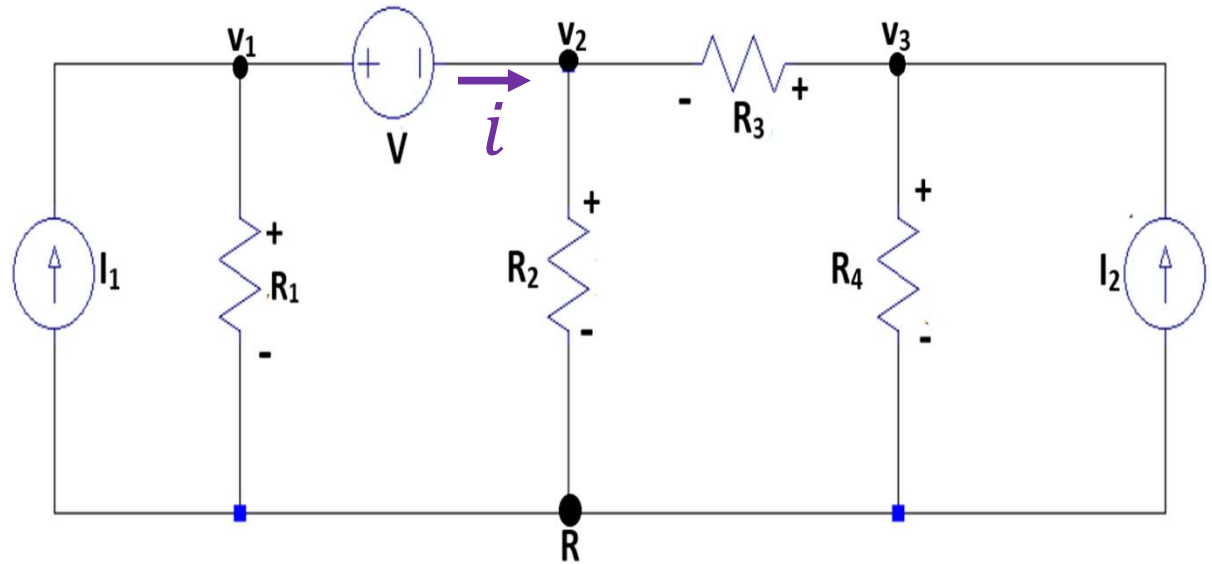
$$-i - \frac{V_1}{R_1} + I_1 = 0$$

$$i + G_1 V_1 = I_1$$

KCL at Node **2**

$$i + \frac{V_3 - V_2}{R_3} - \frac{V_2}{R_2} = 0$$

$$i = (G_2 + G_3)V_2 - G_3V_3$$



$$G_1 V_1 + (G_2 + G_3)V_2 - G_3 V_3 = I_1$$

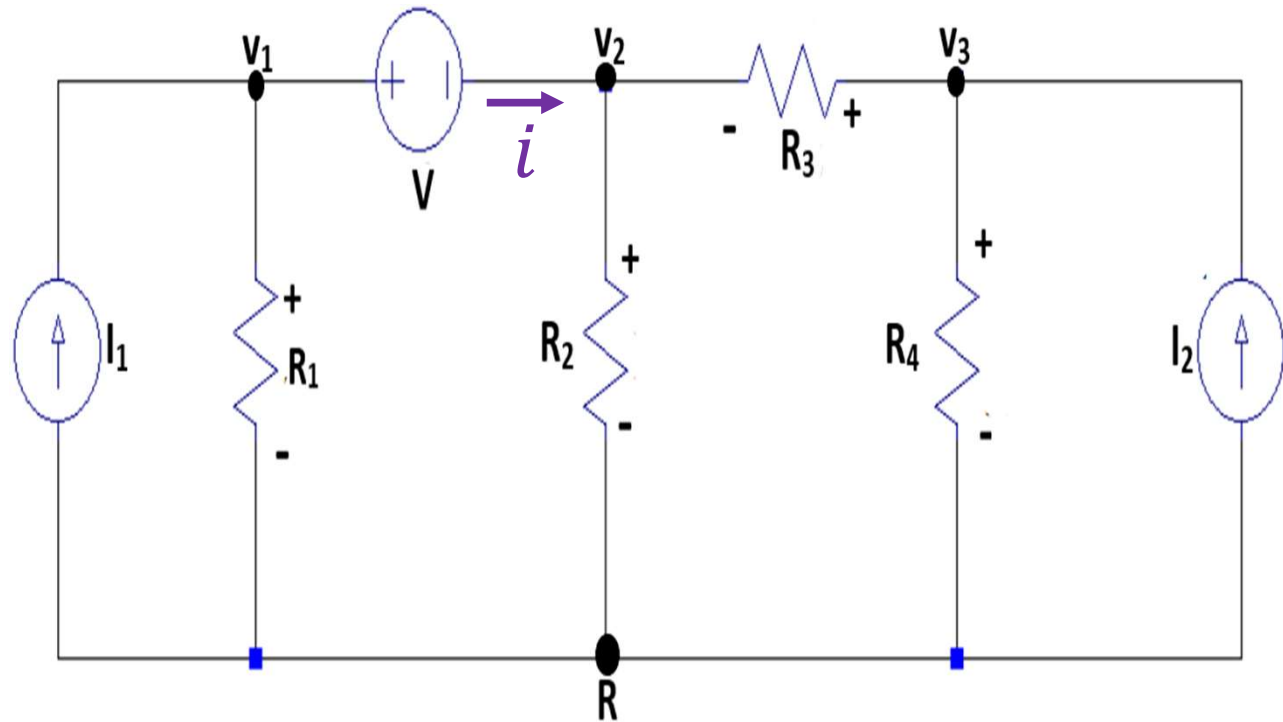
IVS between Two Nodes

KCL at Node 3

$$-\left(\frac{V_3 - V_2}{R_3}\right) - \frac{V_3}{R_4} + I_2 = 0$$
$$-G_3 V_2 + (G_3 + G_4) V_3 = I_2$$

From Nodes 1 & 2

$$V_1 - V_2 = V$$



IVS between Two Nodes

$$G_1 V_1 + (G_2 + G_3) V_2 - G_3 V_3 = I_1$$

$$G_1 (V_2 + V) + (G_2 + G_3) V_2 - G_3 V_3 = I_1$$

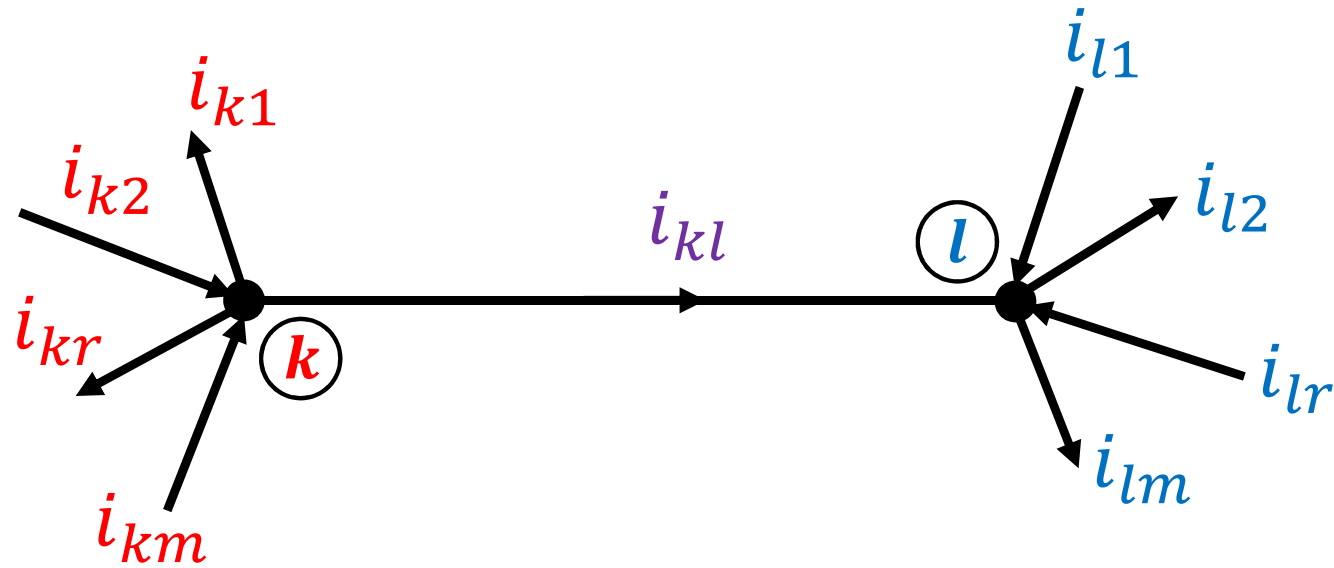
$$(G_1 + G_2 + G_3) V_2 - G_3 V_3 = I_1 - G_1 V$$

$$-G_3 V_2 + (G_3 + G_4) V_3 = I_2$$

$$V_1 - V_2 = V$$

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_3 \\ -G_3 & G_3 + G_4 \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -G_1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ V \end{bmatrix}$$

KCL: An Observation

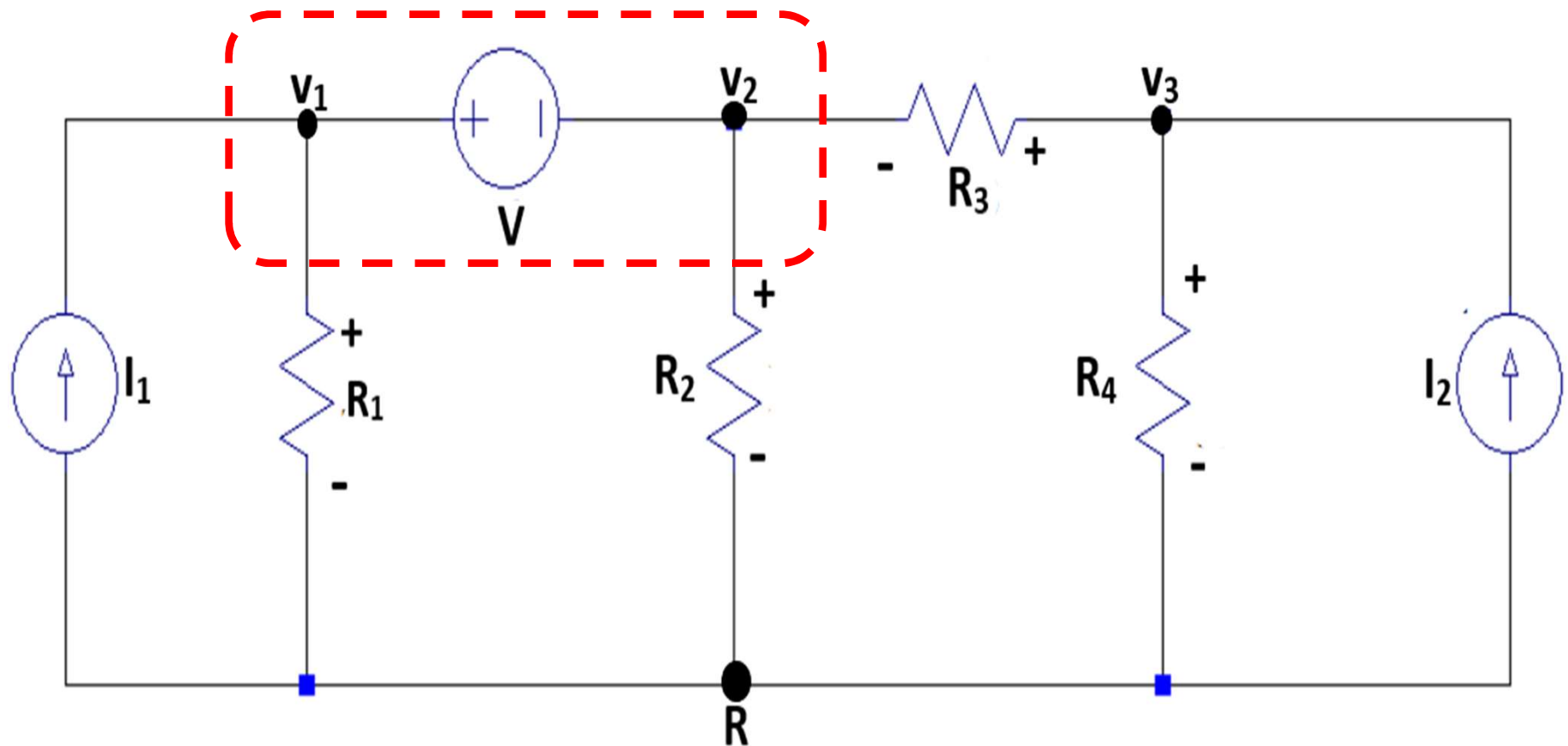


$$-i_{kl} + \sum_{\substack{r=1 \\ r \neq l}}^m i_{kr} = 0$$

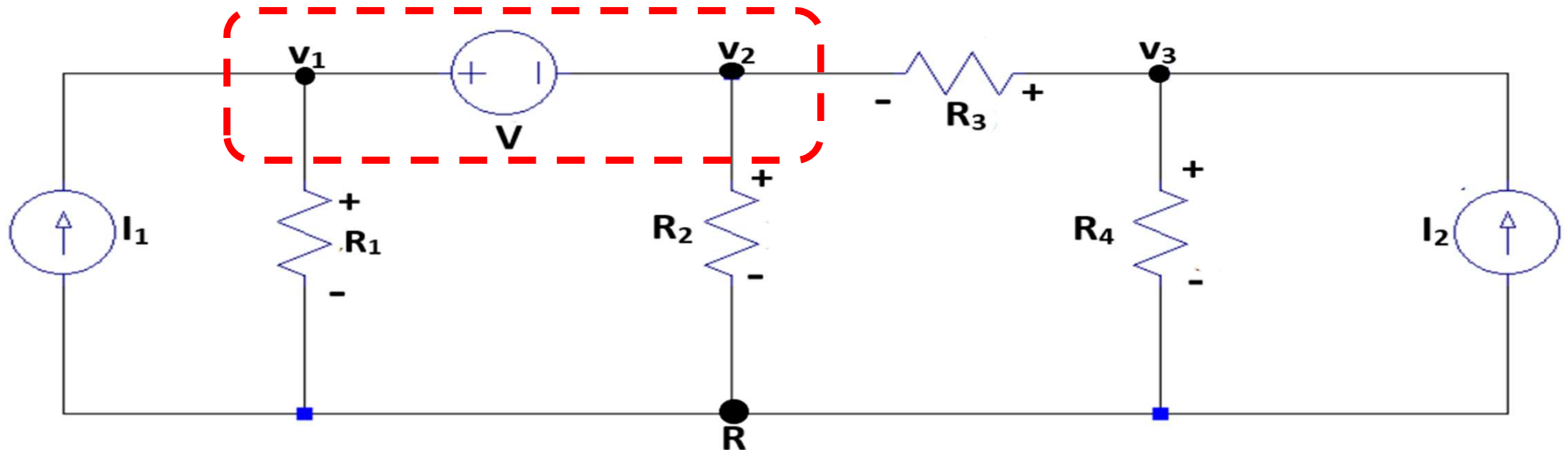
$$i_{kl} + \sum_{\substack{r=1 \\ r \neq k}}^m i_{lr} = 0$$

$$\sum_{\substack{r=1 \\ r \neq l}}^m i_{kr} + \sum_{\substack{r=1 \\ r \neq k}}^m i_{lr} = 0$$

IVS between Two Nodes: Super Node



IVS between Two Nodes: Super Node



$$I_1 + (V_3 - V_2)G_3 = V_1G_1 + V_2G_2$$

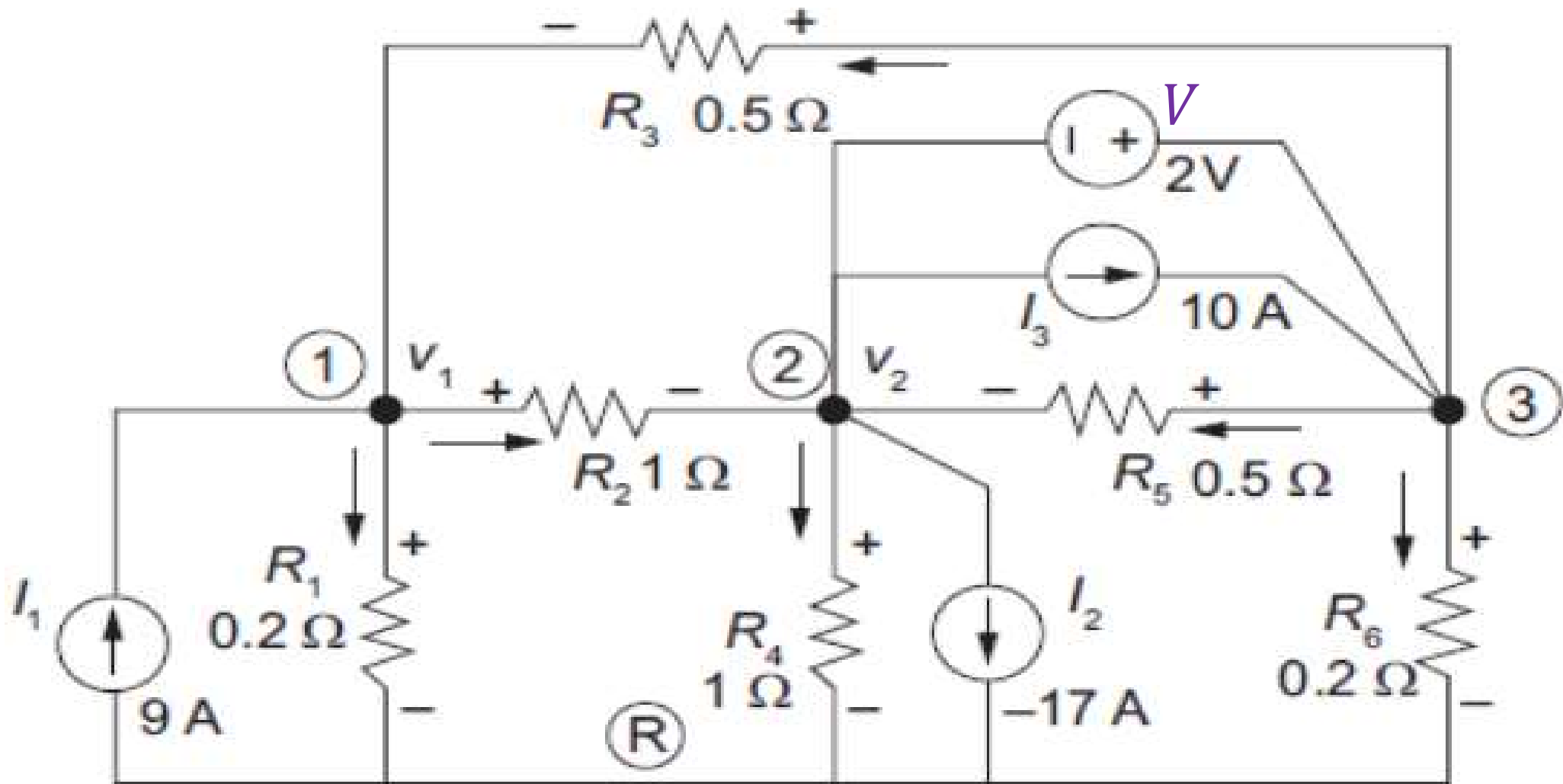
Current Entering
Super Node

Current Leaving
Super Node

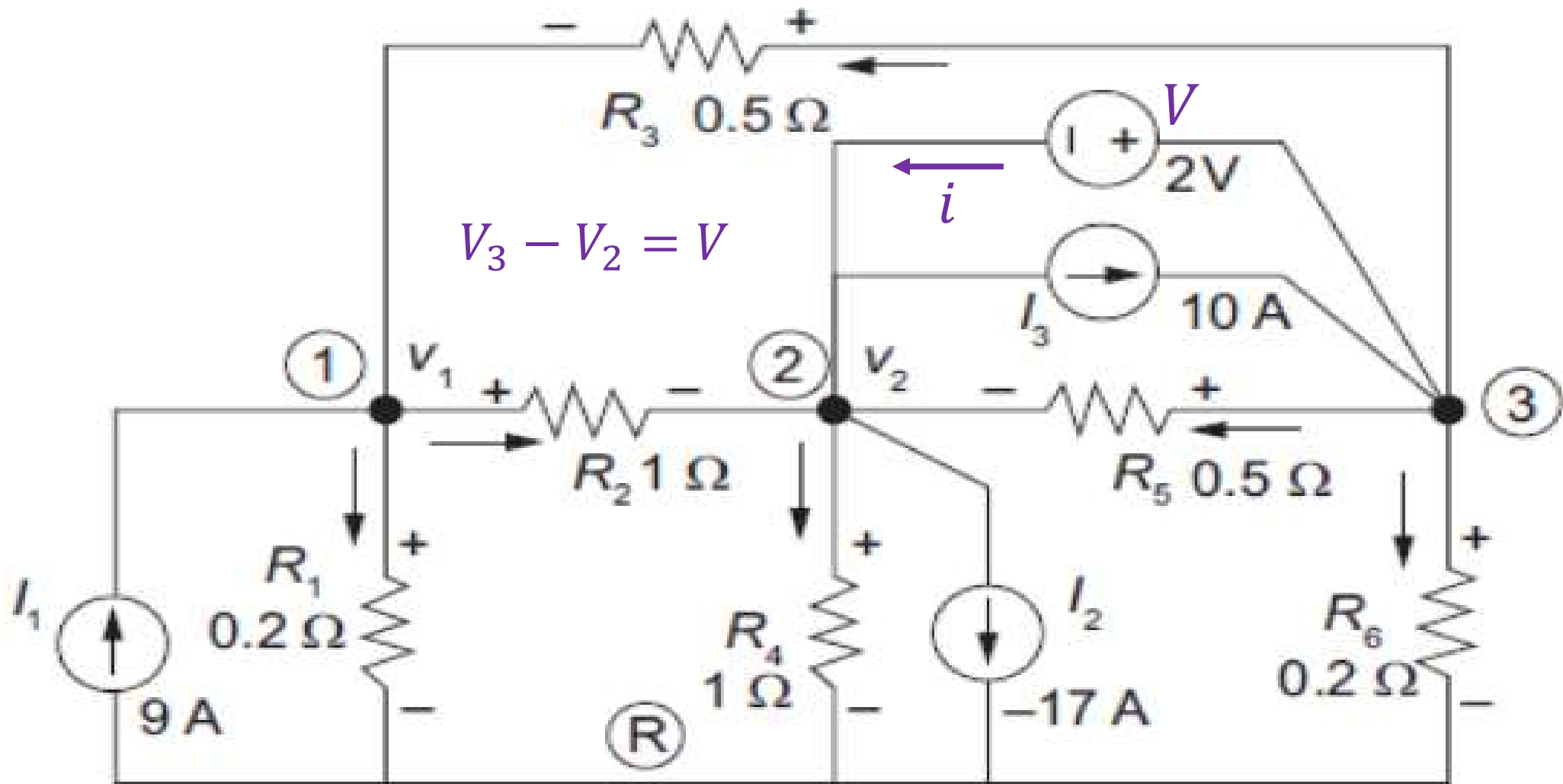
After Rearranging Terms

$$G_1V_1 + (G_2 + G_3)V_2 - G_3V_3 = I_1$$

IVS between Two Nodes



IVS between Two Nodes



IVS between Two Nodes

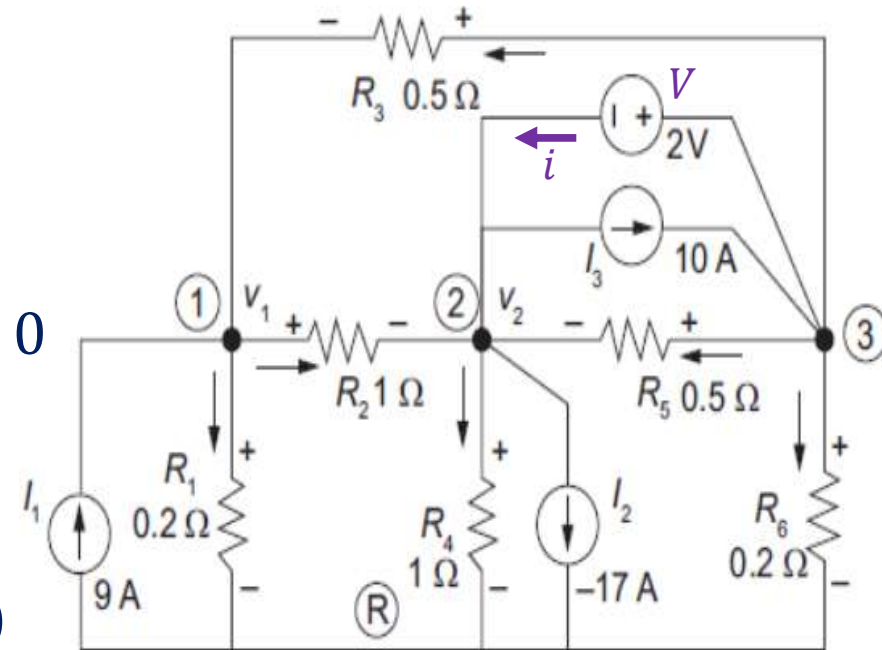
KCL at Node **1**

$$-G_1V_1 - G_2(V_1 - V_2) + I_1 + G_3(V_3 - V_1) = 0$$

$$V_3 - V_2 = V \quad \longrightarrow \quad V_3 = V_2 + V$$

$$G_1V_1 + G_2(V_1 - V_2) = I_1 + G_3(V_2 + V - V_1)$$

$$(G_1 + G_2 + G_3)V_1 - (G_2 + G_3)V_2 = I_1 + G_3V$$



IVS between Two Nodes

KCL at Node **2**

$$G_2(V_1 - V_2) - G_4V_2 - I_2 + G_5V - I_3 + i = 0$$

KCL at Node **3**

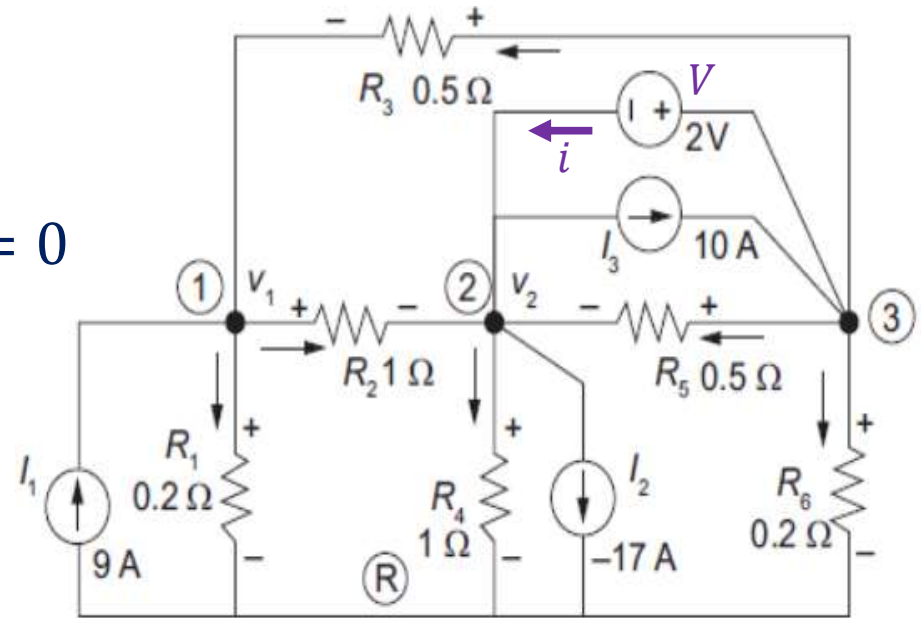
$$I_3 - G_6V_3 - G_5V - i - (V_3 - V_1)G_3 = 0$$

Adding

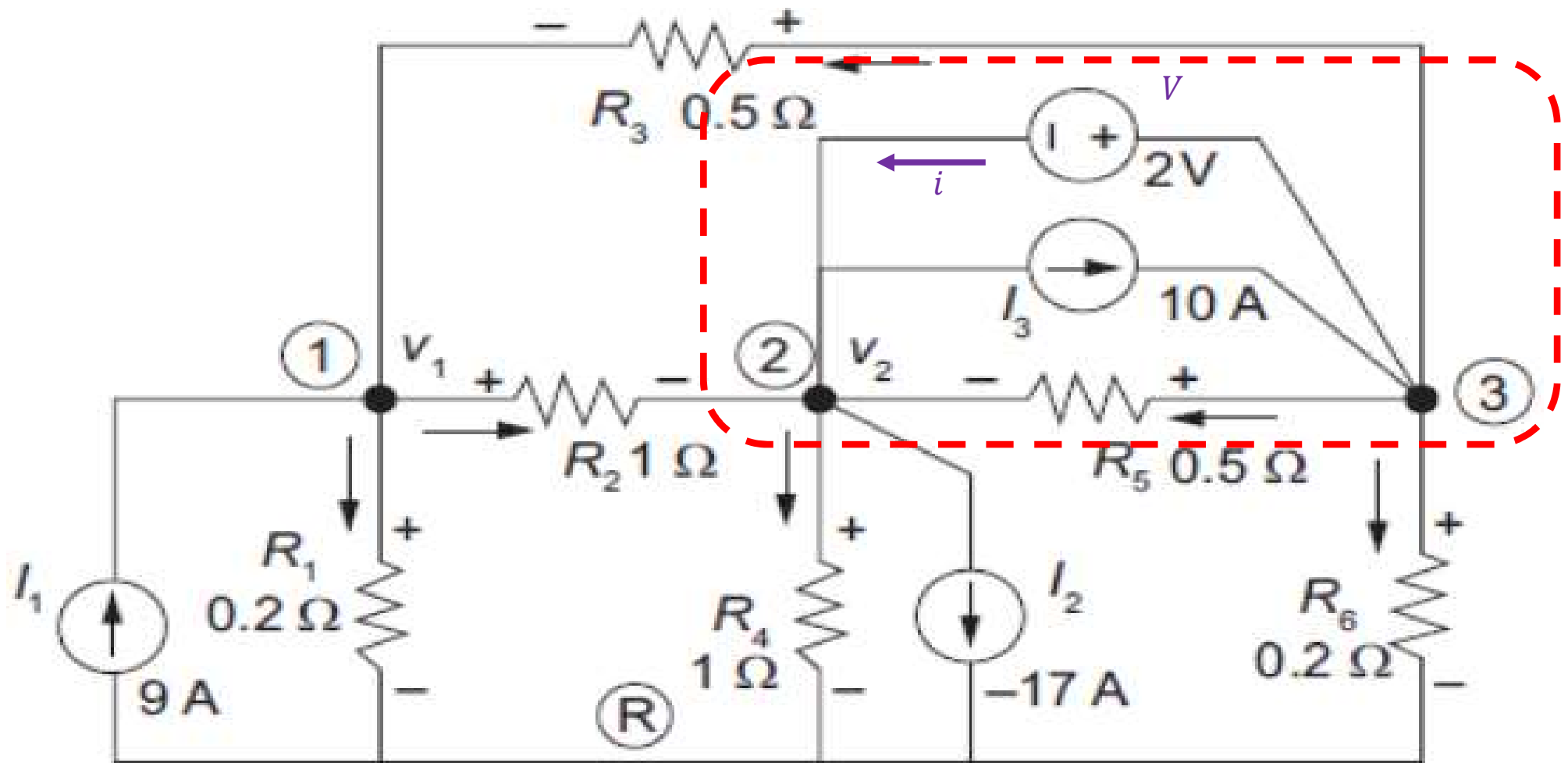
$$G_2(V_1 - V_2) - G_4V_2 - I_2 - G_6V_3 - (V_3 - V_1)G_3 = 0$$

Rearranging ($V_3 = V_2 + V$)

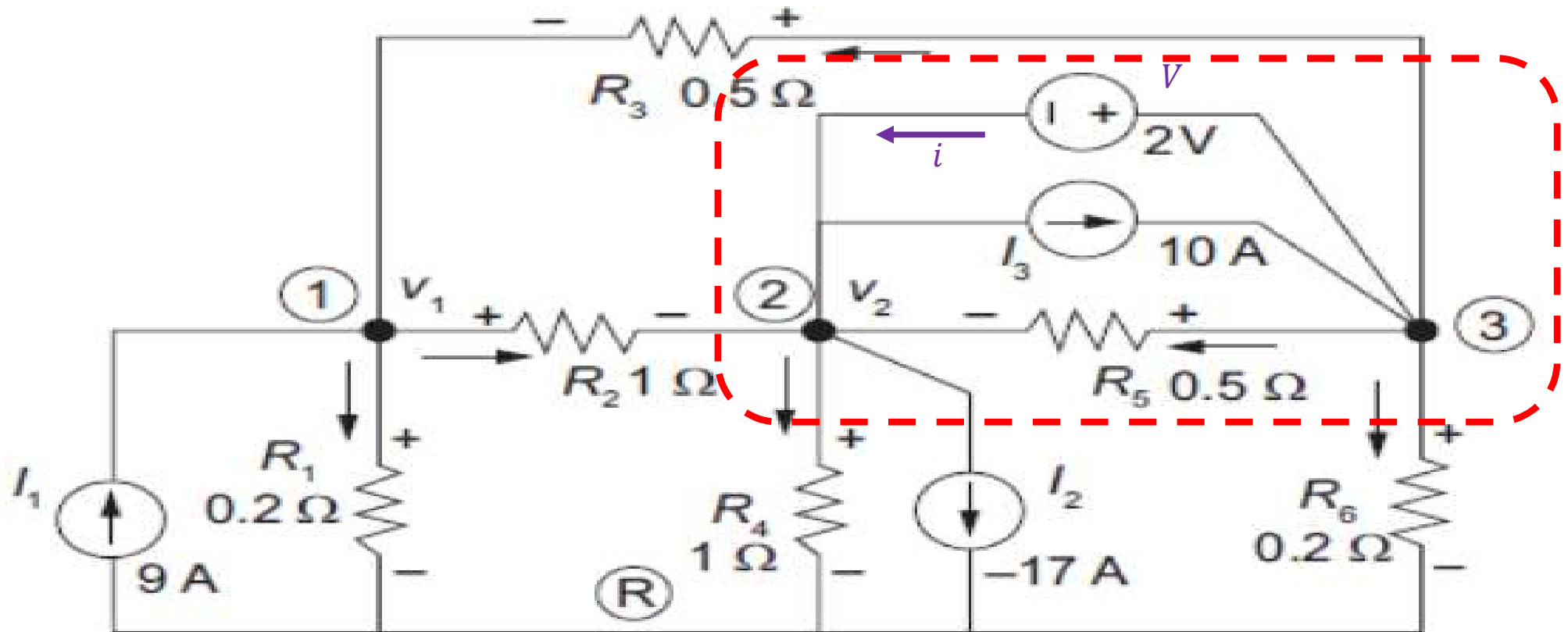
$$-(G_2 + G_3)V_1 + (G_2 + G_3 + G_4 + G_6)V_2 = -I_2 - (G_3 + G_6)V$$



IVS between Two Nodes



IVS between Two Nodes



$$G_2(V_1 - V_2) - G_4V_2 - I_2 - G_6V_3 - G_3(V_3 - V_1) = 0$$

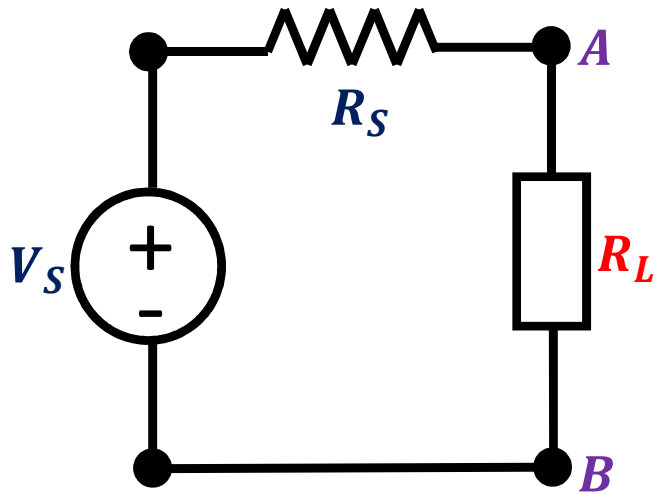
IVS between Two Nodes

$$\left. \begin{aligned} G_2(V_1 - V_2) - G_4V_2 - I_2 - G_6V_3 - G_3(V_3 - V_1) &= 0 \\ G_2(V_1 - V_2) - G_4V_2 - I_2 - G_6(V_2 + V) - G_3(V_2 + V - V_1) &= 0 \\ -(G_2 + G_3)V_1 + (G_2 + G_3 + G_4 + G_6)V_2 &= -I_2 - (G_3 + G_6)V \end{aligned} \right\} V_3 = V_2 + V$$

$$(G_1 + G_2 + G_3)V_1 - (G_2 + G_3)V_2 = I_1 + G_3V$$

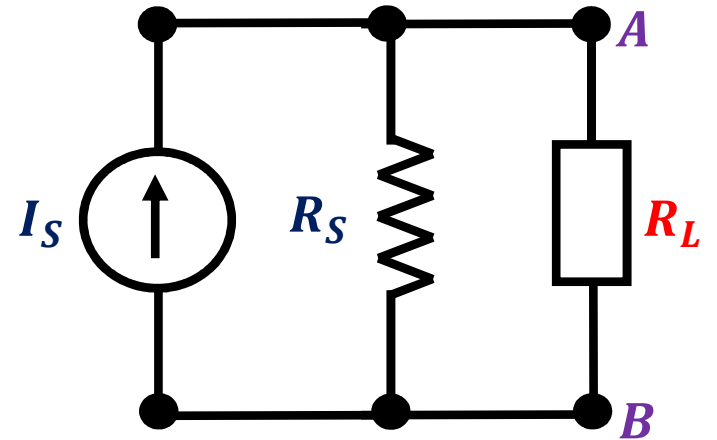
$$\begin{bmatrix} G_1 + G_2 + G_3 & -(G_2 + G_3) \\ -(G_2 + G_3) & G_2 + G_3 + G_4 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & G_3 \\ 0 & -1 & 0 & -(G_3 + G_6) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V \end{bmatrix}$$

Source Transformation



$$V_{AB}(V_S) = ?$$

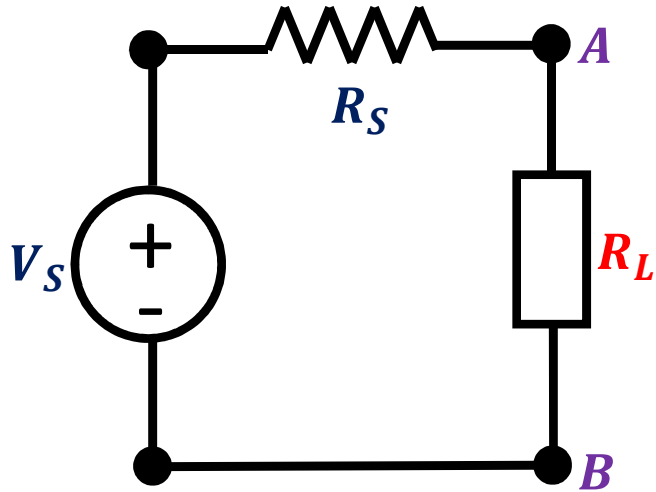
$$I_{AB}(V_S) = ?$$



$$V_{AB}(I_S) = ?$$

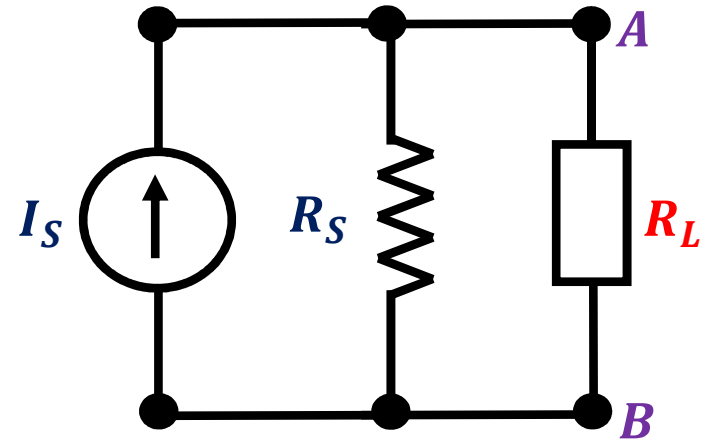
$$I_{AB}(I_S) = ?$$

Source Transformation



$$V_{AB}(V_S) = \frac{V_S R_L}{R_S + R_L}$$

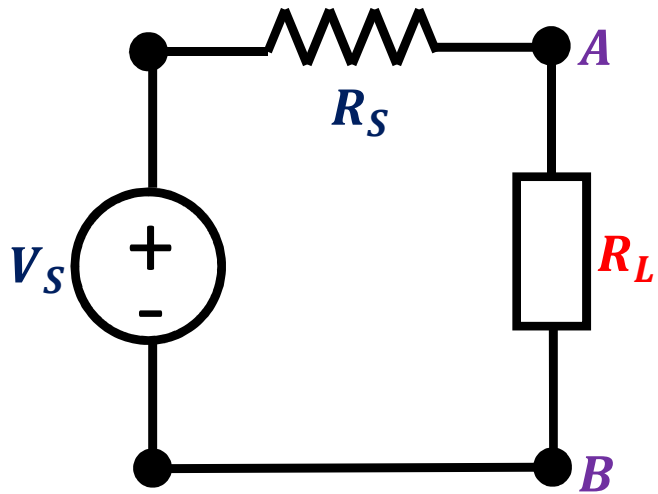
$$I_{AB}(V_S) = \frac{V_S}{R_S + R_L}$$



$$V_{AB}(I_S) = \frac{I_S R_S R_L}{R_S + R_L}$$

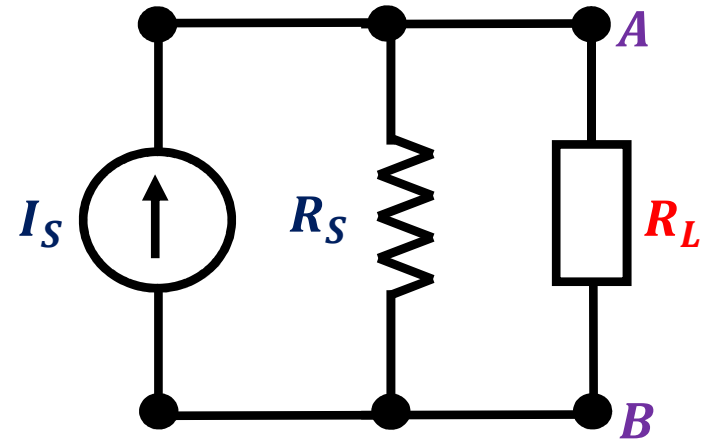
$$I_{AB}(I_S) = \frac{I_S R_S}{R_S + R_L}$$

Source Transformation



$$V_{AB}(V_S) = V_{AB}(I_S)$$

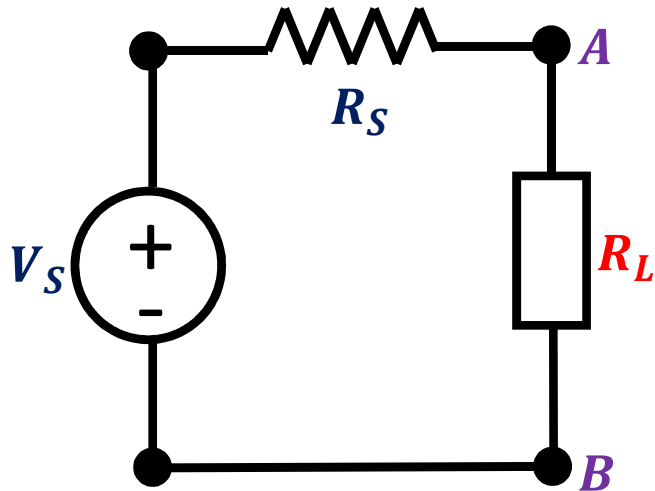
$$I_{AB}(V_S) = I_{AB}(I_S)$$



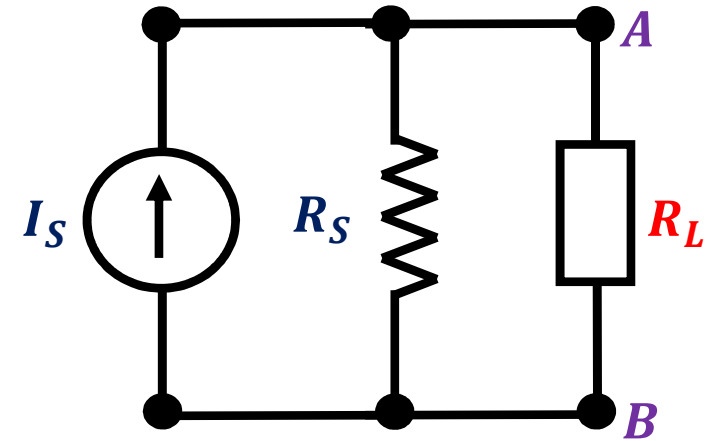
$$I_S = ?$$

Source Transformation Theorem

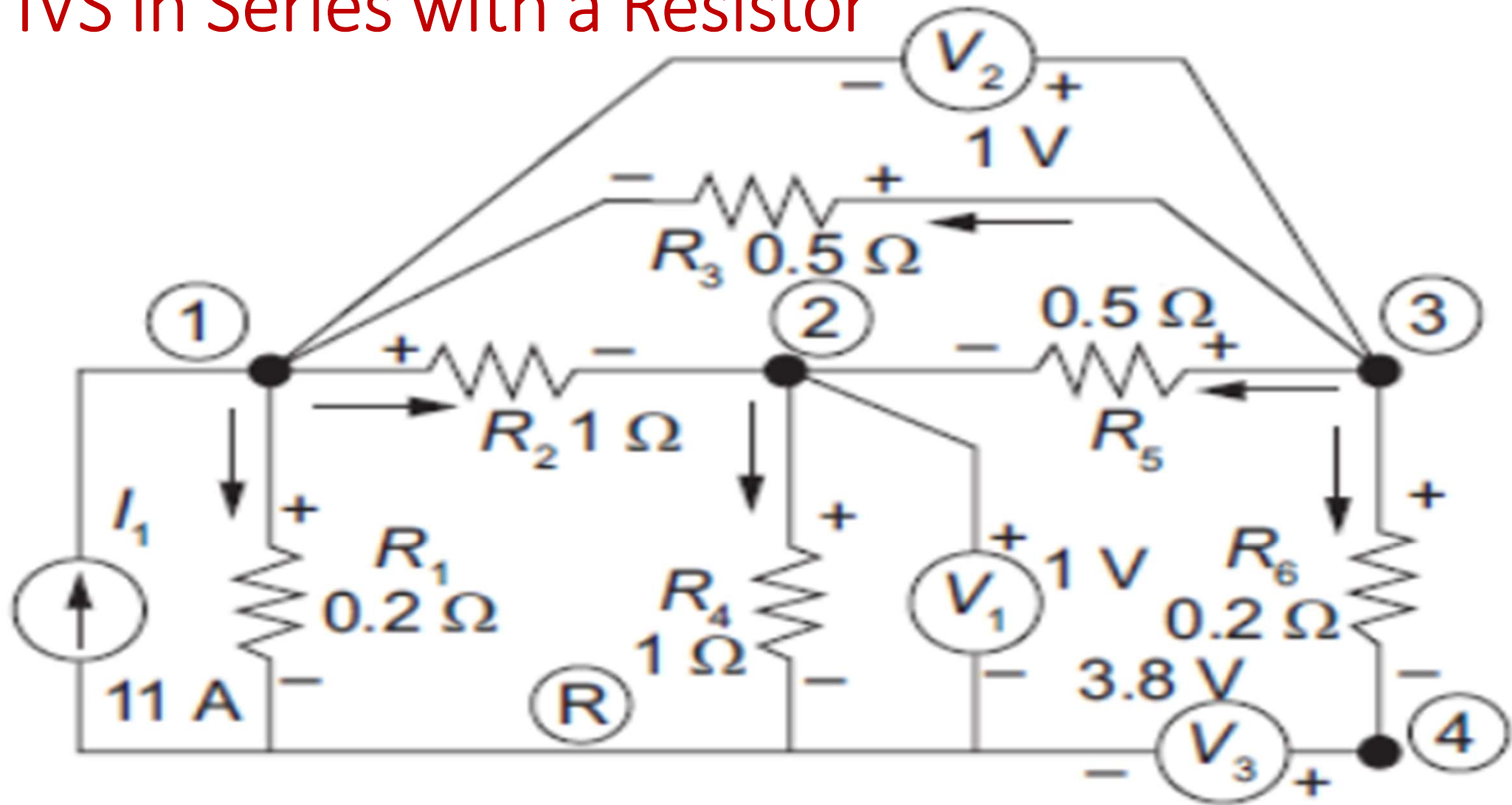
A Voltage Source with Source Function V_S in Series with a Resistance R_S can be Replaced by a Current Source with Source Function I_S in Parallel with a Resistance R_S without affecting any Voltage or Current or Power variable external to the source. The direction of the Current Source is such that Current Flows Out of the Terminal at which the Positive of the Voltage Source is Presently Connected and Vice versa.



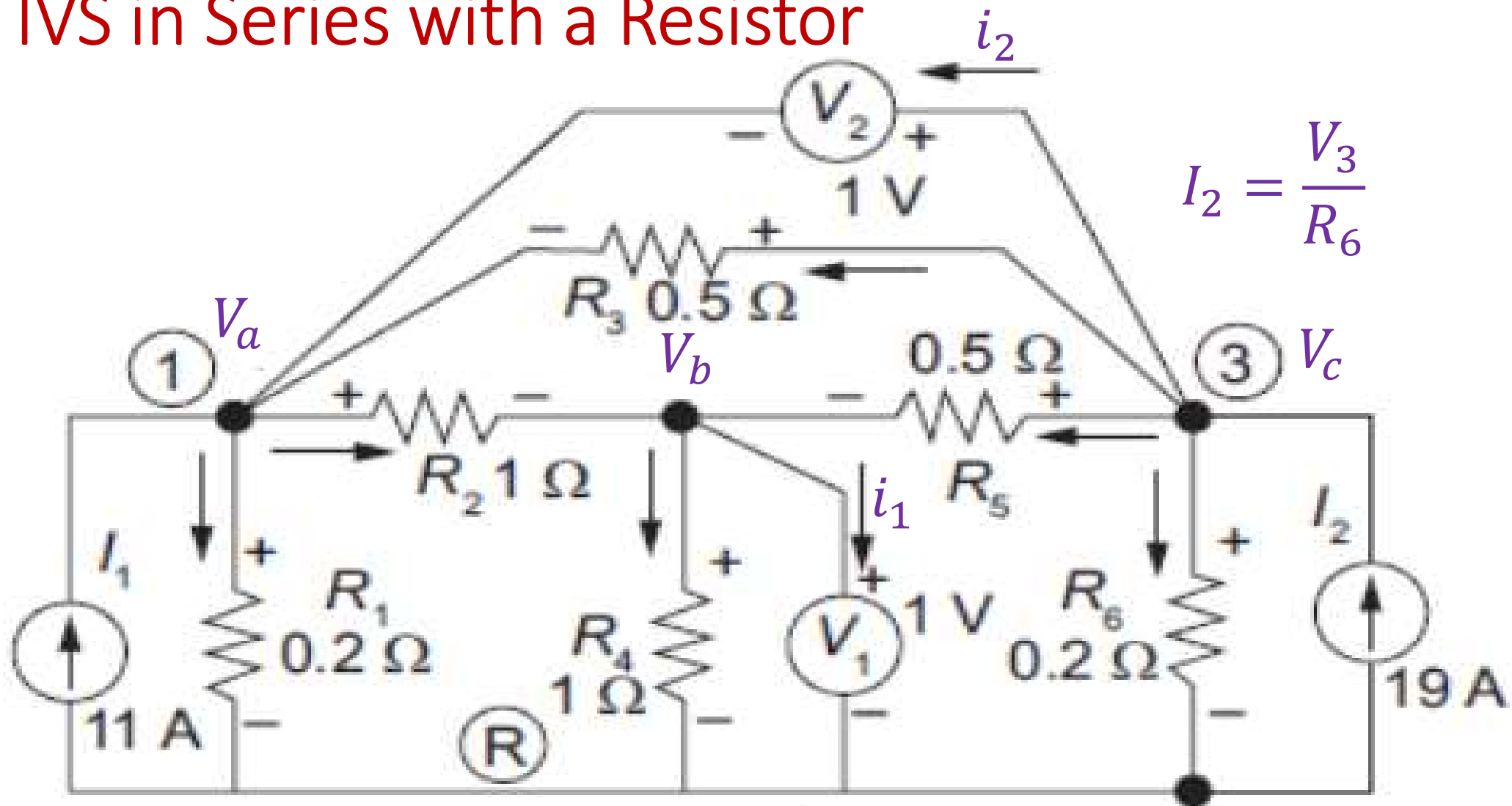
$$I_S = \frac{V_S}{R_S}$$



IVS in Series with a Resistor



IVS in Series with a Resistor



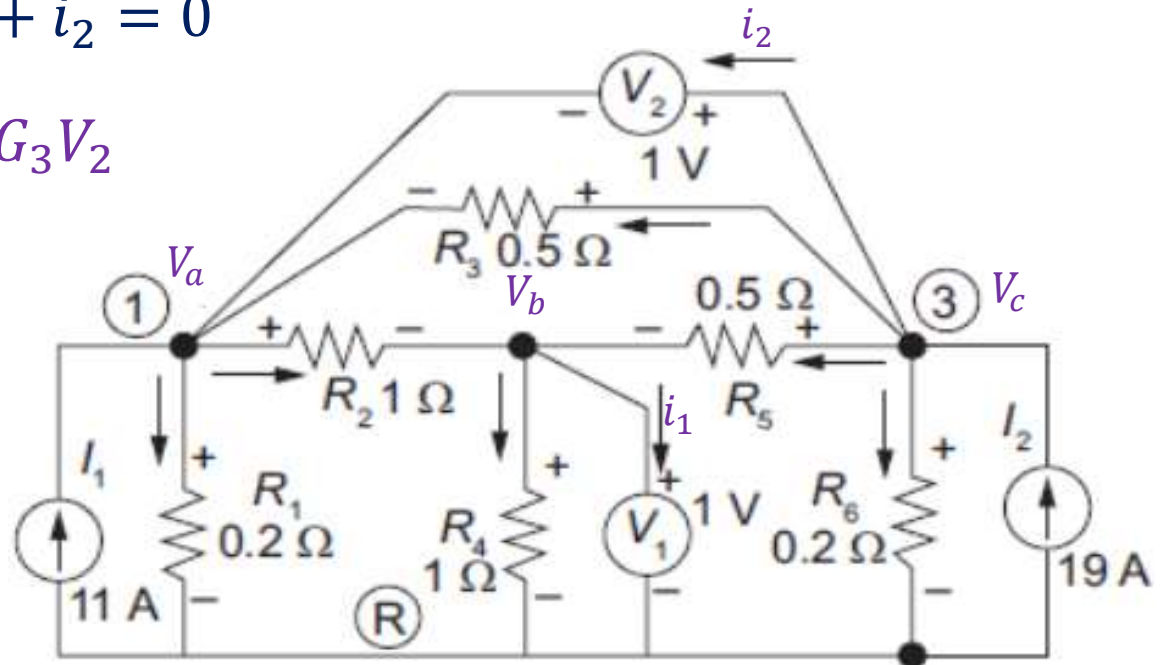
IVS in Series with a Resistor

KCL at Node **1**

$$I_1 - G_1 V_a - G_2(V_a - V_b) + G_3(V_c - V_a) + i_2 = 0$$

$$I_1 - G_1 V_a - G_2(V_a - V_1) + G_3 V_2 + i_2 = 0$$

$$(G_1 + G_2)V_a - i_2 = I_1 + G_2 V_1 + G_3 V_2$$



$$V_b = V_1$$

$$V_c = V_a + V_2$$

IVS in Series with a Resistor

KCL at Node **3**

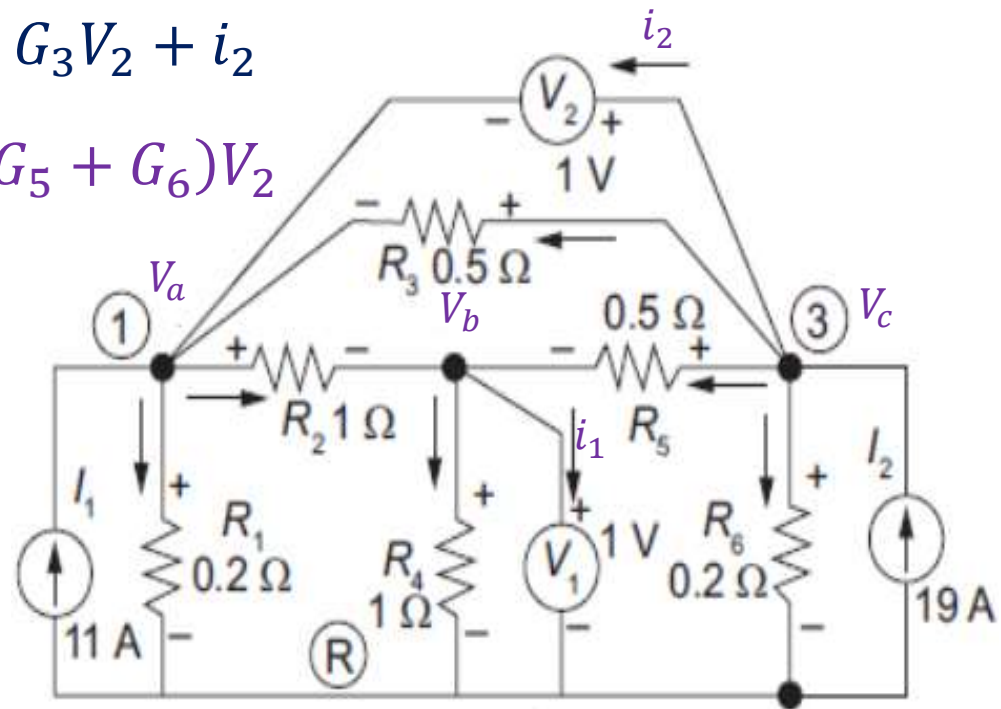
$$I_2 = G_6 V_c + G_5(V_c - V_b) + G_3(V_c - V_a) + i_2$$

$$I_2 = G_6(V_a + V_2) + G_5(V_a + V_2 - V_b) + G_3 V_2 + i_2$$

$$(G_5 + G_6)V_a + i_2 = I_2 + G_5 V_1 - (G_3 + G_5 + G_6)V_2$$

$$V_b = V_1$$

$$V_c = V_a + V_2$$



IVS in Series with a Resistor

$$(G_1 + G_2)V_a - i_2 = I_1 + G_2V_1 + G_3V_2$$

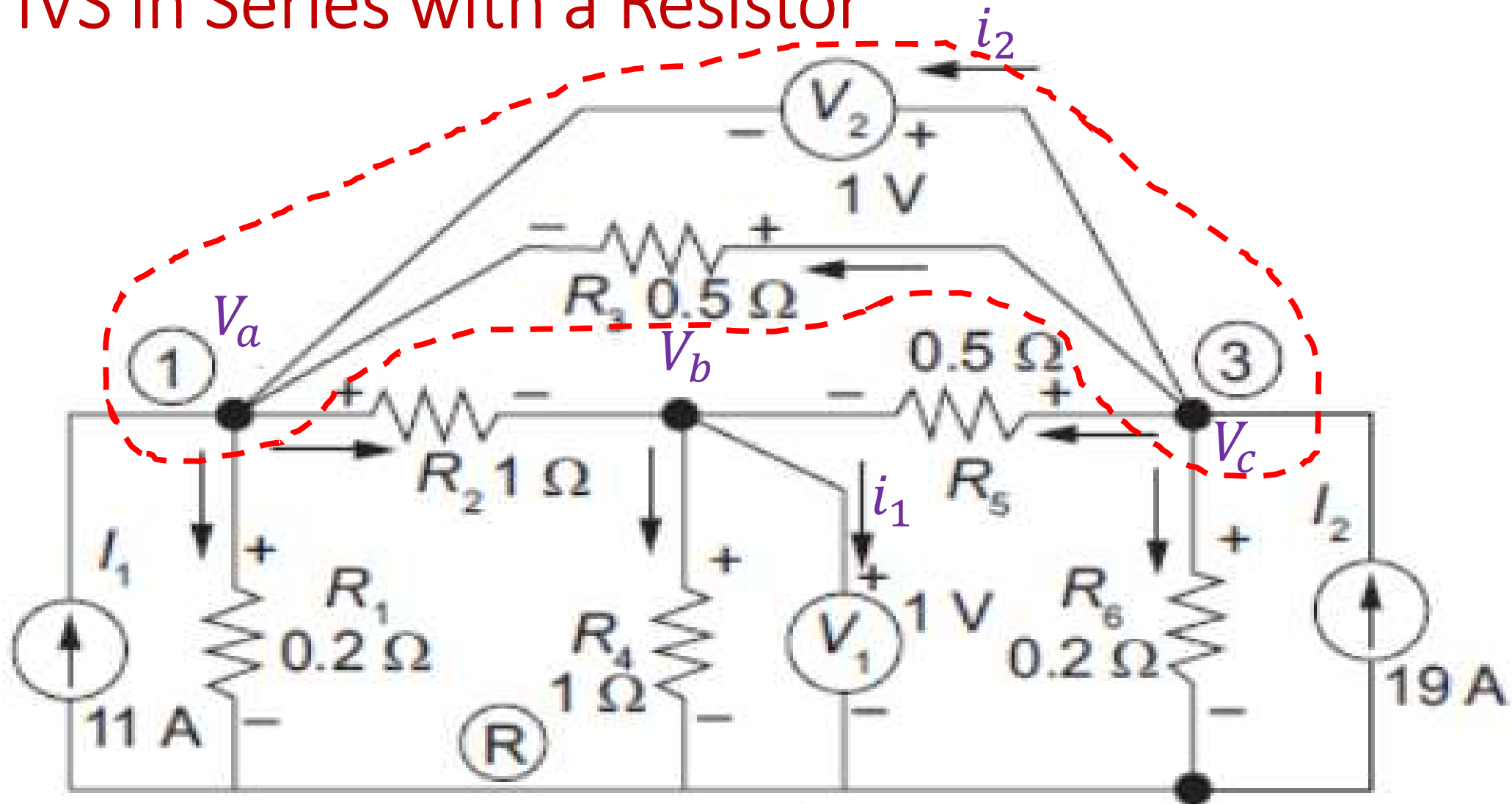
$$(G_5 + G_6)V_a + i_2 = I_2 + G_5V_1 - (G_3 + G_5 + G_6)V_2$$

$$(G_1 + G_2 + G_5 + G_6)V_a = I_1 + I_2 + (G_2 + G_5)V_1 - (G_5 + G_6)V_2$$

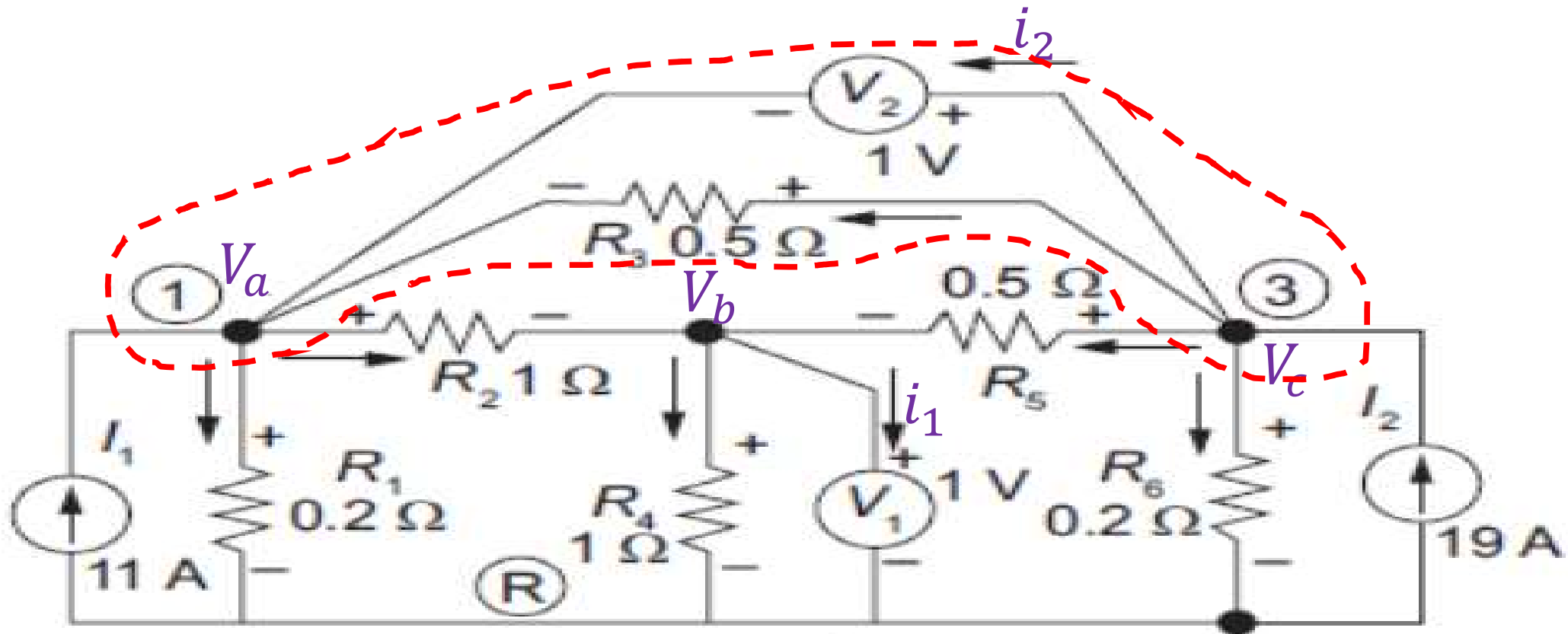
Adding

$$[G_1 + G_2 + G_5 + G_6][V_a] = [1 \quad G_2 + G_5 \quad -(G_5 + G_6) \quad G_6] \begin{bmatrix} I_1 \\ V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

IVS in Series with a Resistor



IVS in Series with a Resistor



$$I_1 - G_1 V_a - G_2 (V_a - V_b) - G_5 (V_c - V_b) - G_6 V_c + I_2 = 0$$

IVS in Series with a Resistor

$$I_1 - G_1 V_a - G_2(V_a - V_b) - G_5(V_c - V_b) - G_6 V_c + I_2 = 0$$

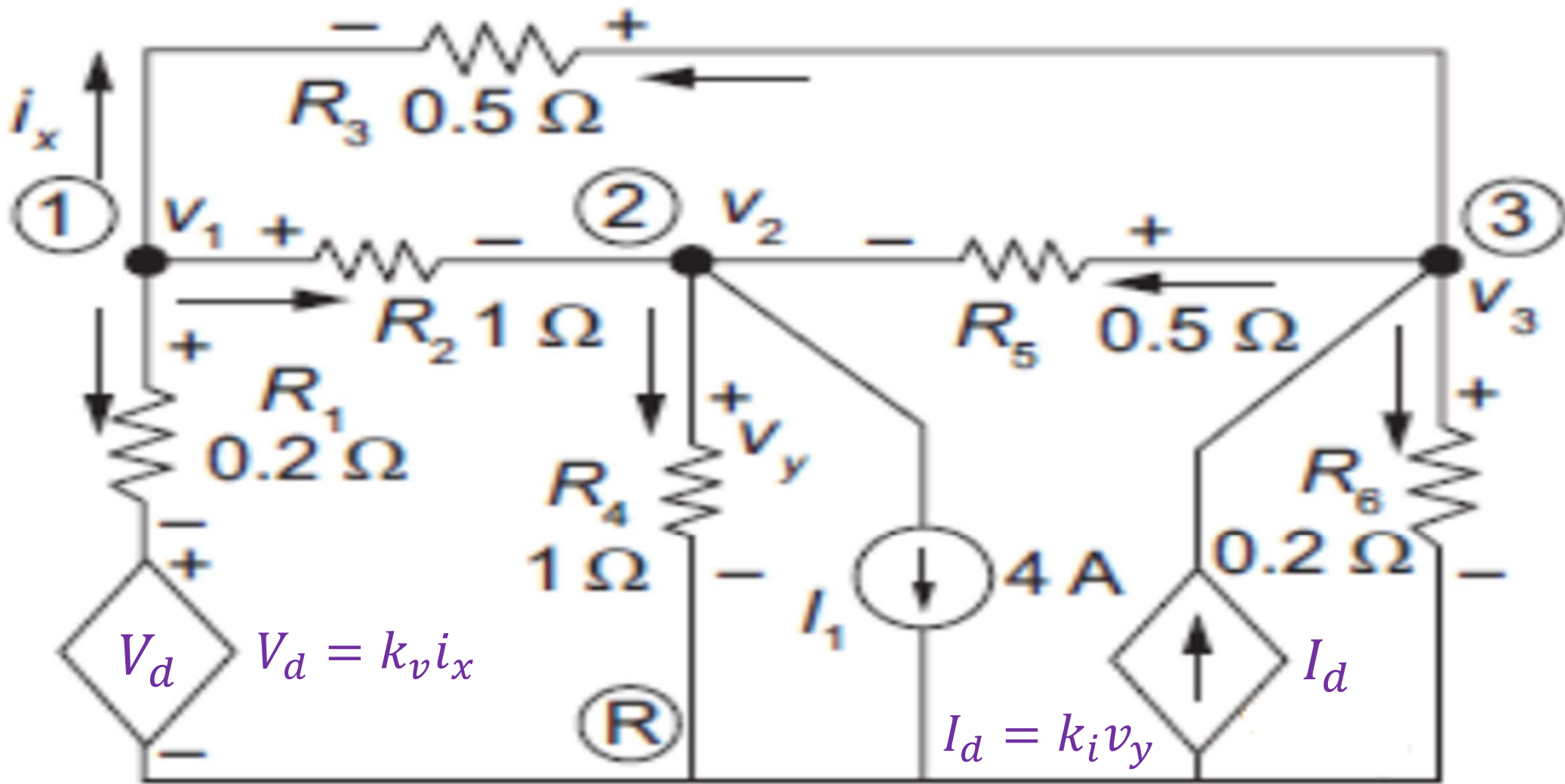
$$V_b = V_1 \quad \downarrow \quad V_c = V_a + V_2$$

$$I_1 - G_1 V_a - G_2(V_a - V_1) - G_5(V_a + V_2 - V_1) - G_6(V_a + V_2) + I_2 = 0$$

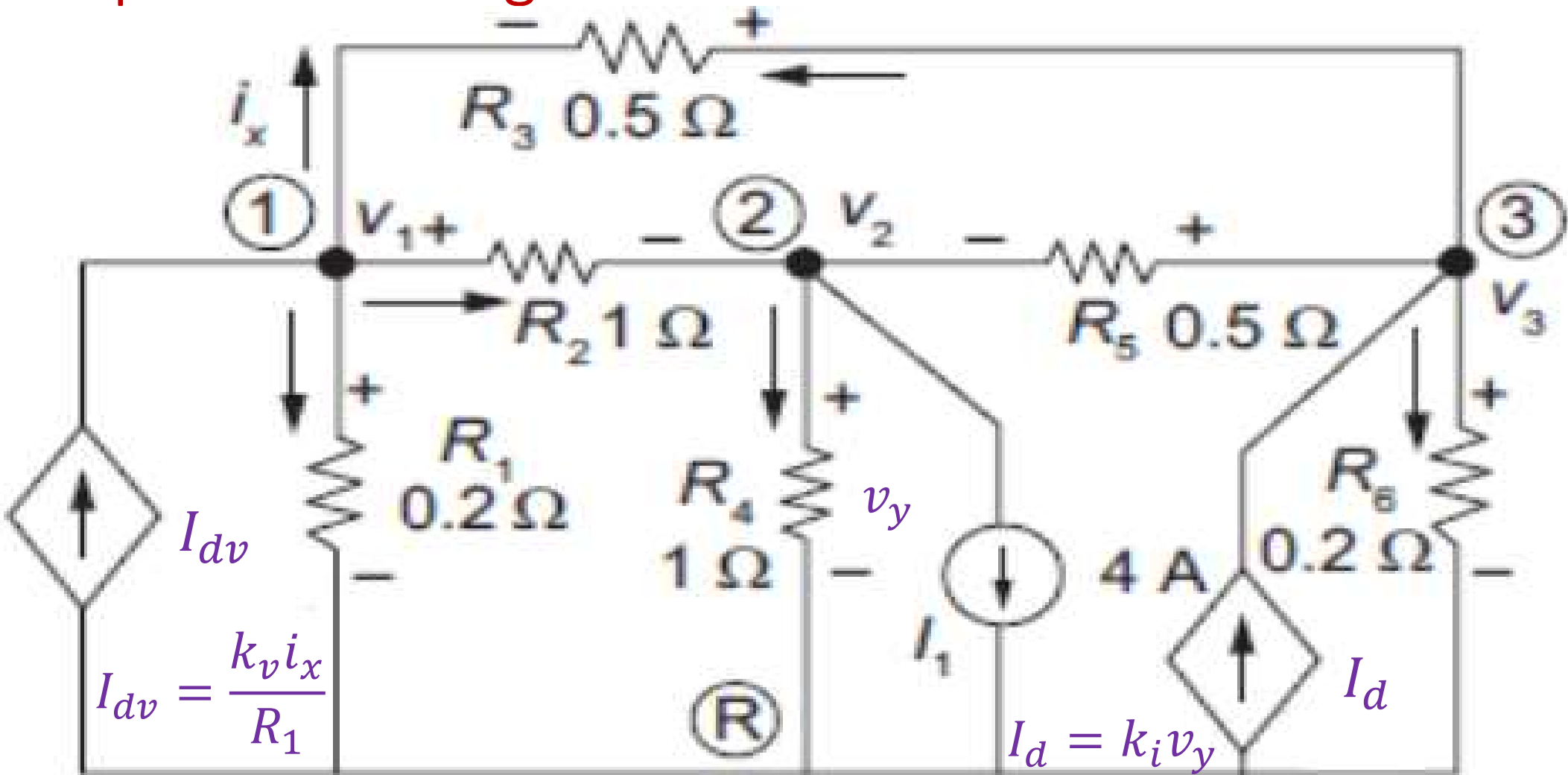
$$I_1 + I_2 - (G_1 + G_2 + G_5 + G_6)V_a + (G_2 + G_5)V_1 - (G_5 + G_6)V_2 = 0$$

$$(G_1 + G_2 + G_5 + G_6)V_a = I_1 + (G_2 + G_5)V_1 - (G_5 + G_6)V_2 + G_6 V_3$$

Dependent Voltage and Current Source



Dependent Voltage and Current Source



Dependent Voltage and Current Source

KCL at Node **1**

$$I_{dv} - G_1 V_1 - G_2 (V_1 - V_2) + G_3 (V_3 - V_1) = 0$$

$$k_v i_x G_1 - (G_1 + G_2 + G_3) V_1 + G_2 V_2 + G_3 V_3 = 0$$

$$k_v \{G_3 (V_1 - V_3)\} G_1 - (G_1 + G_2 + G_3) V_1 + G_2 V_2 + G_3 V_3 = 0$$

$$(G_1 + G_2 + G_3 - k_v G_1 G_3) V_1 - G_2 V_2 - (G_3 - k_v G_1 G_3) V_3 = 0$$

KCL at Node **2**

$$G_2 (V_1 - V_2) - G_4 V_2 - I_1 + G_5 (V_3 - V_2) = 0$$

$$-G_2 V_1 + (G_2 + G_4 + G_5) V_2 - G_5 V_3 = -I_1$$

Dependent Voltage and Current Source

KCL at Node **3**

$$-G_3(V_3 - V_1) - G_5(V_3 - V_2) + I_d - G_6V_3 = 0$$

$$G_3V_1 + G_5V_2 - (G_3 + G_5 + G_6)V_3 + k_iV_2 = 0$$

$$-G_3V_1 - (G_5 + k_i)V_2 + (G_3 + G_5 + G_6)V_3 = 0$$

$$\begin{bmatrix} G_1 + G_2 + G_3 - k_v G_1 G_3 & -G_2 & -(G_3 - k_v G_1 G_3) \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -(G_5 + k_i) & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} [I_1]$$

Summary

- Nodal Analysis with Current and Voltage Sources
- IVS between Node and Reference Node
- IVS between Two Non-reference Nodes
- Concept of Super Node
- IVS in Series with a Resistor
- Nodal Analysis with Dependent Sources



Thank You